



Joint Model Importance via the Rashomon Set

Behavioural Comparison of Near-Equally-Good Models

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Beyond the single best model

- In tabular ML, many models reach indistinguishable predictive performance.
- For a tolerance $\varepsilon > 0$, the *Rashomon set*

$$\text{RS}(\varepsilon) = \{ \hat{f}_{\mathcal{D},\lambda} : \mathcal{R}(\hat{f}_{\mathcal{D},\lambda}) \leq \mathcal{R}(\hat{f}_{\mathcal{D},\lambda^*}) \cdot (1 + \varepsilon), \lambda \in \Lambda \}$$

collects every model within ε of the best.

Notation: $\hat{f}_{\mathcal{D},\lambda}$ = model trained on data $\mathcal{D}$ with hyperparameter configuration $\lambda \in \Lambda$; λ^* = risk-minimizing configuration; $\mathcal{R}(\cdot)$ = risk.

- Members of $\text{RS}(\varepsilon)$ can disagree substantially on individual predictions and on feature importance.
- **Question.** How are these equally good models related to one another, and what does the collective tell us beyond the single best?

- 1 For every $\hat{f}_{\mathcal{D},\lambda} \in \text{RS}(\varepsilon)$, compute the prediction vector $\hat{y}$ on a held-out 2/3 validation split.
- 2 Build the pairwise *behavioural distance* matrix D with entries

$$d_{ij} = \|\hat{y}_i - \hat{y}_j\|_p = \left(\sum_{o=1}^{n_{\text{val}}} |\hat{y}_{i,o} - \hat{y}_{j,o}|^p \right)^{1/p},$$

where i, j index models in $\text{RS}(\varepsilon)$ and o runs over the n_{val} validation observations.

- 3 View D three ways:
 - ▶ hierarchical clustering (`hclust`)
 - ▶ classical MDS (`cmdscale`, embedding dimension $k = 2$)
 - ▶ partitioning around medoids (`cluster::pam`)

Distances computed for both Euclidean ($p = 2$) and Manhattan ($p = 1$); results shown are Euclidean (Manhattan tells the same story).

Rashomon tolerance $\varepsilon = 1\%$ of the minimum risk.

Five learner families considered for every task: `tree`, `glmnet`, `xgb`, `nnet`, `svm`.

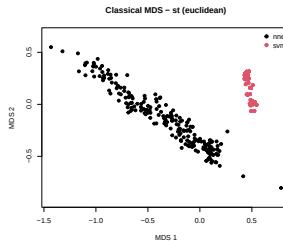
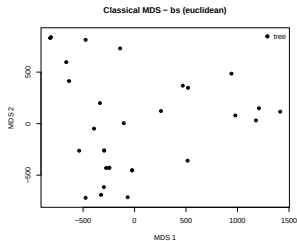
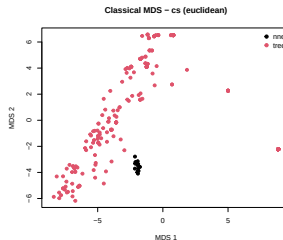
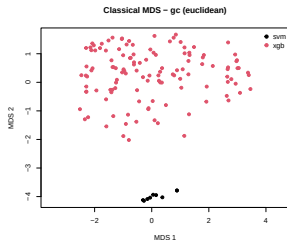
Key	Task	Type
gc	German Credit	classification
cs	COMPAS	classification
bs	Bike Sharing	regression
st	Synthetic	regression

Who survives the Rashomon cut?

Task	Surviving families (n models)
gc	svm(9), xgb(125)
cs	nnet(15), tree(261)
bs	tree(30)
st	nnet(216), svm(160)

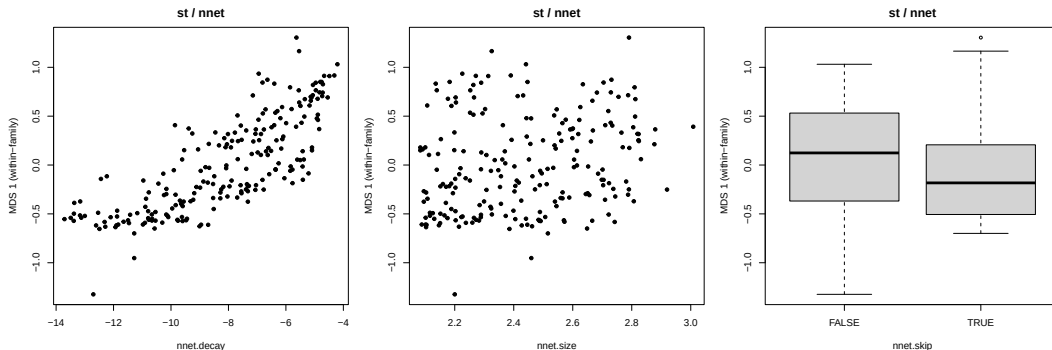
First finding. At $\varepsilon = 1\%$ only one or two learner families survive per task — *learner choice* is the first dimension of disagreement inside $RS(\varepsilon)$.

MDS view — all four tasks



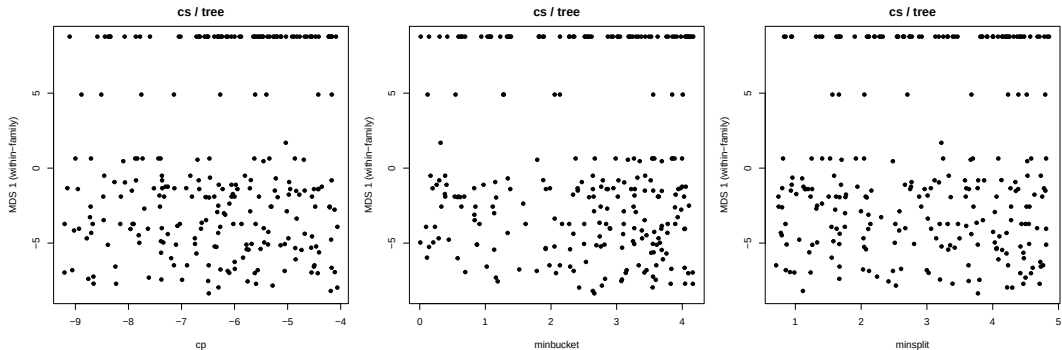
- Wherever two families coexist, behavioural distance **separates them cleanly** — the dominant axis is family identity, not within-family variation.
- Within-family variance is *asymmetric*: `svm` and `nnet` form tight clusters; `tree` and `xgb` scatter widely despite fewer tunable hyperparameters.
- Several families trace out a **1D-manifold** in the MDS plane (e.g. `st/nnet`, `cs/tree`) — next two slides.

Within-family structure I — st / nnet



- Each panel: 1D within-family MDS coordinate vs. one hyperparameter.
- The within-family manifold **is the decay axis** — near-monotone mapping.
- size and skip have no effect on behavioural position.

Within-family structure II — cs / tree

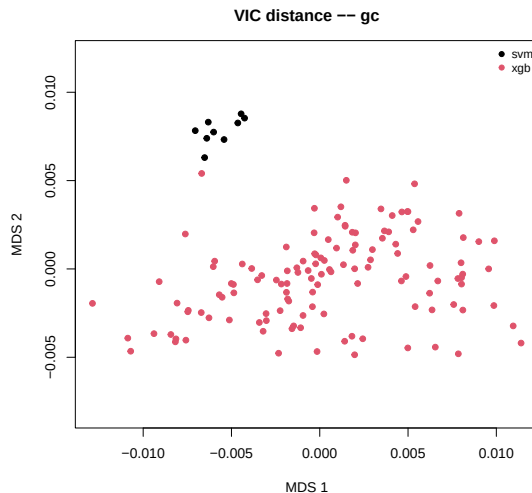
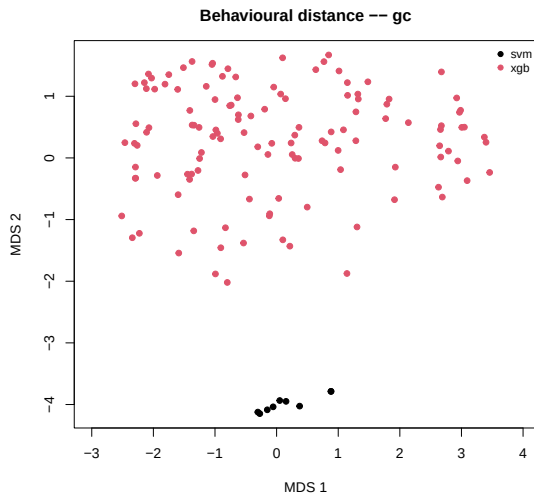


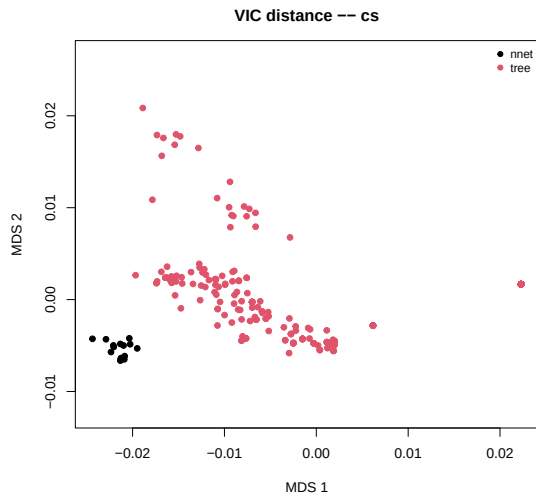
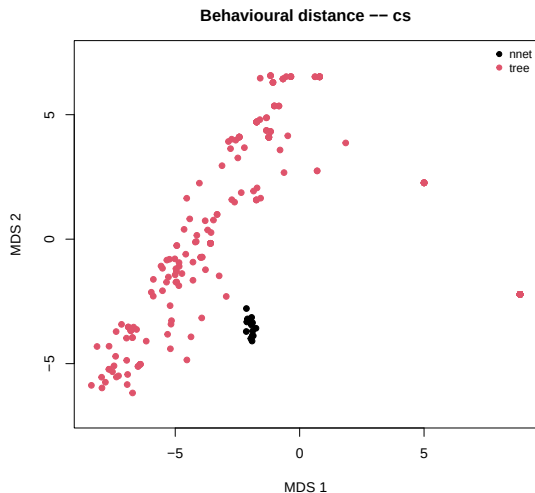
- Within-family MDS coordinate vs. cp, minbucket, minsplit.
- Trees on COMPAS are **bimodal**: a tight cluster at $\text{MDS } 1 \approx 8$ and a diffuse population below zero.
- *No single tuned hyperparameter* separates the two modes. **Speculation**: a discrete behavioural split driven by structure (e.g. root split, depth) that hyperparameter tuning does not reach — to be confirmed.

Scope — behaviour is not importance

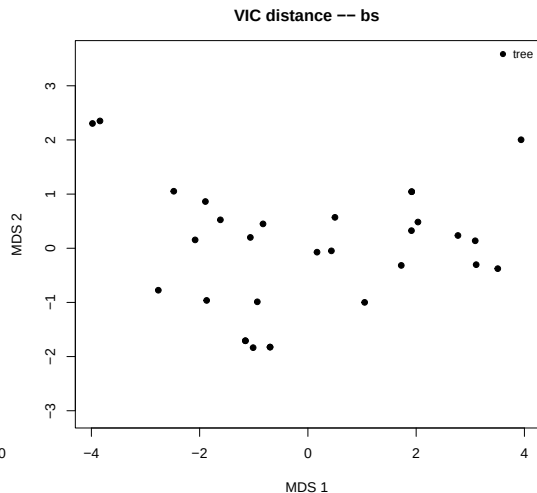
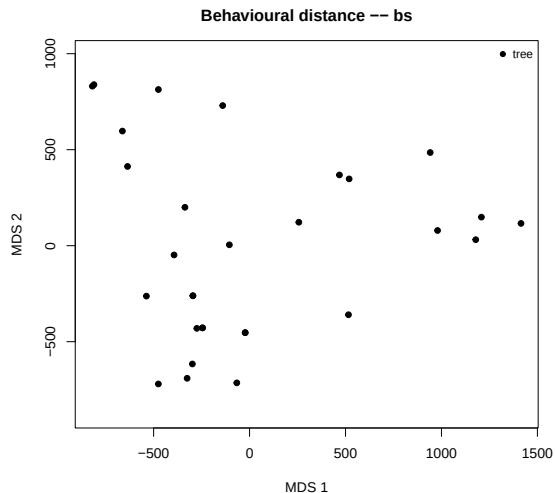
- **What we measure:** behavioural distance $d_{ij}^{\text{beh}} = \|\hat{y}_i - \hat{y}_j\|_p$ from validation predictions.
- **What “joint model importance” asks:** how do per-model importance vectors $\phi^{(i)}$ vary across $i \in \text{RS}(\varepsilon)$? (Variable-Importance Cloud, VIC.)
- **Independent in principle:** identical predictions need not imply identical importance (and vice versa).
- **Compared empirically next slide:** same models, same colouring, behavioural distance left, VIC distance right.

Behavioural vs VIC — gc

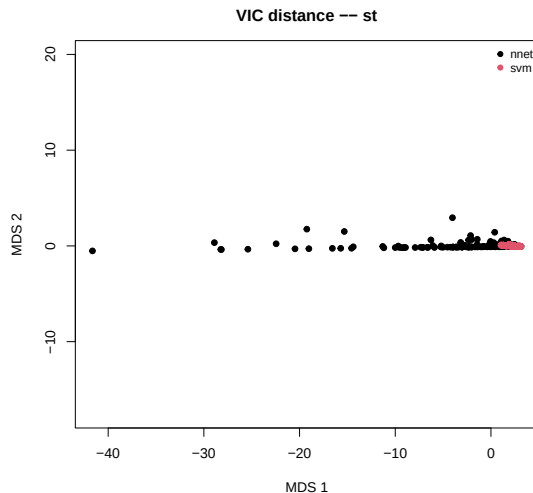
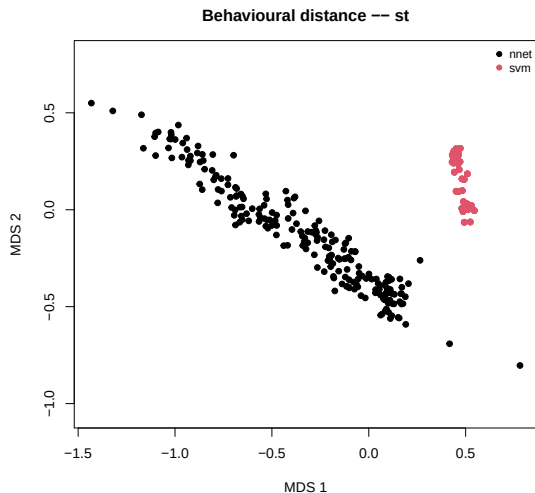




Behavioural vs VIC — bs



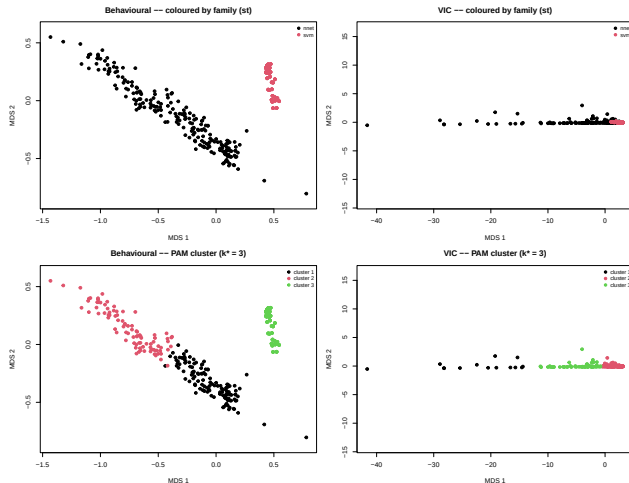
Behavioural vs VIC — st



Behavioural vs VIC — what we see

- gc: family separation largely preserved in both views, though a handful of xgb points overlap the svm cluster in VIC.
- cs: family separation preserved, but the within-family behavioural bimodality of trees (Slide 9) **vanishes** on the importance side.
- bs: single family in $RS(\varepsilon)$ — uninformative.
- st: behavioural family separation **collapses** in VIC — nnet and svm share the same importance profile despite predicting very differently.
- **Take-away:** behavioural clustering is *not* a reliable proxy for importance clustering. The disagreement is empirical and *task-dependent*.

Letting PAM choose k freely — st



- Silhouette-tuned PAM picks $k^* = 3$ for st in *both* views, not 2.
- **Behavioural:** PAM splits the nnet manifold along the decay axis into two clusters — the within-family structure is discrete enough to read as its own cluster.
- **VIC:** a different $k^* = 3$ that cuts *across* the family colouring — importance has its own organisation, not just a collapsed copy of behaviour.
- Family identity is therefore neither necessary nor sufficient to recover the unsupervised cluster structure.

First interpretation

- Within $RS(\varepsilon)$, **behaviour has two-level structure**: learner family + position on the family's behavioural manifold.
- Across every task observed, family choice dominates behavioural variance — not hyperparameter setting.
- The within-family manifold can be smooth and HP-aligned ($st/nnet \sim \text{decay}$) or discrete and HP-orthogonal ($cs/tree$; *causal mechanism still speculative*).
- **But behaviour is not importance**: family separation only partly survives in VIC (gc , cs), and the within-family discreteness of $cs/tree$ vanishes entirely. For st , the families themselves collapse. *Joint model importance therefore requires both views.*

- **ε sensitivity.** How do behavioural and VIC clusters shift as the Rashomon tolerance opens?
- **Per-feature analysis.** Which features drive the behaviour-vs-VIC disagreement? Especially for the st collapse.
- **Medoids as representatives.** Treat the silhouette-tuned PAM medoids as a compact reduction of $RS(\varepsilon)$ for downstream interpretation.
- **Disagreement map.** Where in feature space do Rashomon-set models disagree the most?
- **Discrete tree mode on cs.** Is the behavioural bimodality driven by structural differences (root split, depth) that hyperparameters do not capture?

Thank you.

Questions?

Backup — the behavioural distance matrix in detail

Stack the Rashomon-set prediction vectors row-wise into a prediction matrix:

$$\mathbf{P} = \begin{pmatrix} \hat{y}_{1,1} & \hat{y}_{1,2} & \cdots & \hat{y}_{1,n_{\text{val}}} \\ \hat{y}_{2,1} & \hat{y}_{2,2} & \cdots & \hat{y}_{2,n_{\text{val}}} \\ \vdots & \vdots & & \vdots \\ \hat{y}_{n,1} & \hat{y}_{n,2} & \cdots & \hat{y}_{n,n_{\text{val}}} \end{pmatrix} \in \mathbb{R}^{n \times n_{\text{val}}}$$

with one row per model in $\text{RS}(\varepsilon)$ (here $n = |\text{RS}(\varepsilon)|$) and one column per held-out validation observation (here n_{val}).

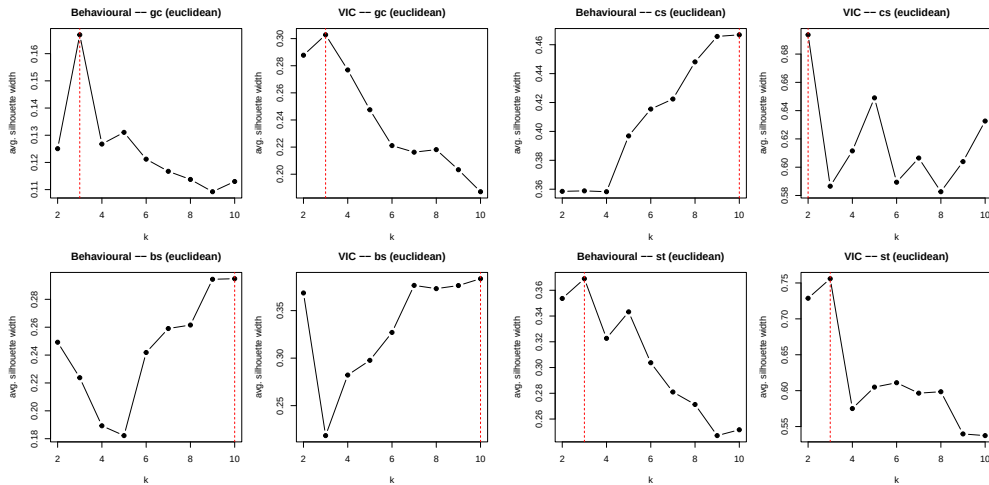
Row-wise L_p distances collapse the observation axis and yield the symmetric matrix

$$\mathbf{D} \in \mathbb{R}^{n \times n}, \quad d_{ij} = \|\hat{y}_i - \hat{y}_j\|_p.$$

Rows / columns of $\mathbf{D}$ now index *models*, not observations. Hence **one MDS point = one model**; the cloud sits in $\mathbb{R}^n$ and is visualised in $\mathbb{R}^2$ via `cmdscale`.

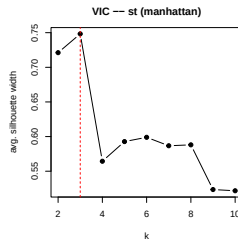
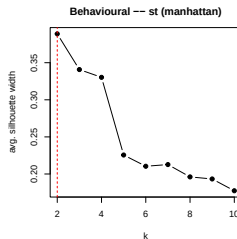
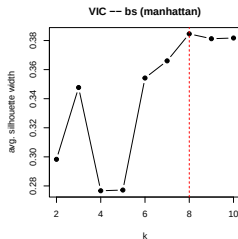
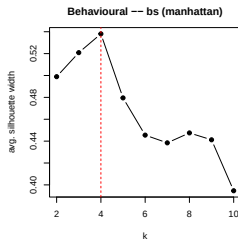
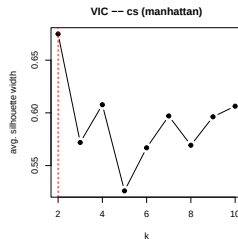
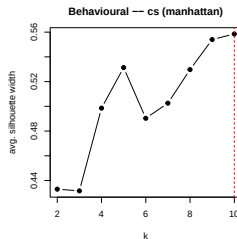
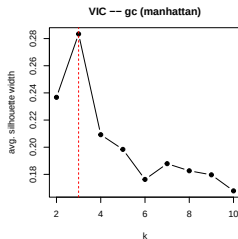
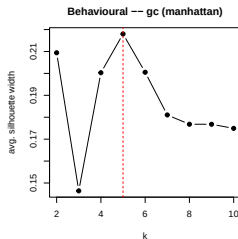
- **Hierarchical clustering** (`hclust`): bottom-up greedy merging of the two closest clusters; output is a dendrogram. Cluster count is fixed by where one cuts.
- **Classical MDS** (`cmdscale`, $k = 2$): eigendecomposition of the double-centred squared distance matrix, top- k eigenvectors as coordinates. Each model becomes a point in $\mathbb{R}^k$ whose pairwise Euclidean distances approximate D. For Euclidean D **equivalent to PCA** up to sign and rotation.
- **Partitioning around medoids** (`cluster::pam`): pick k medoids (representative models) minimising the total distance from each model to its nearest medoid. Robust analogue of k -means for arbitrary distance matrices; best k via average silhouette width.

Backup — silhouette scans (Euclidean)

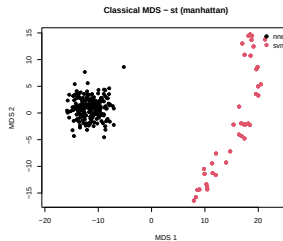
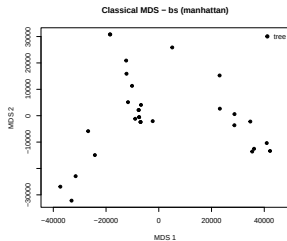
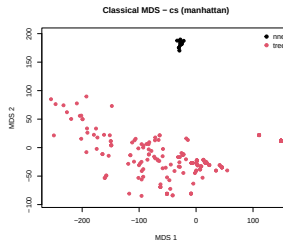
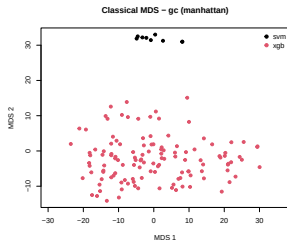


Avg. silhouette width over $k = 2, \dots, 10$. Behavioural (left) and VIC (right) per task; red dashed line marks k^* .

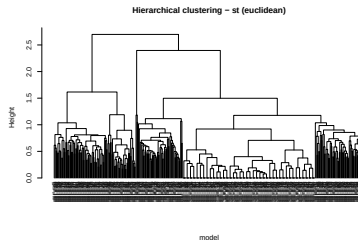
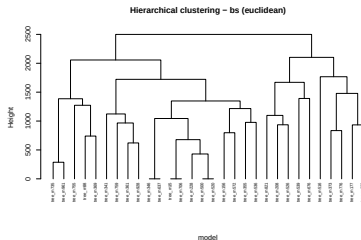
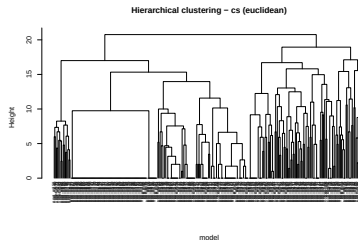
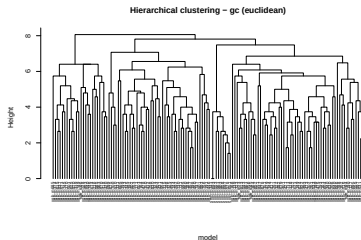
Backup — silhouette scans (Manhattan)



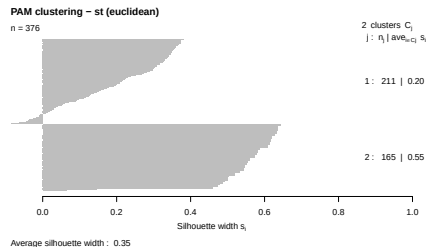
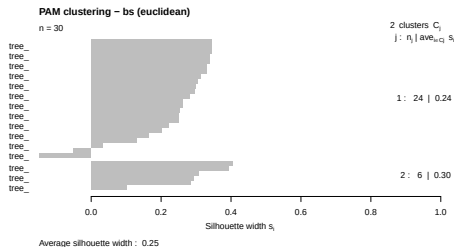
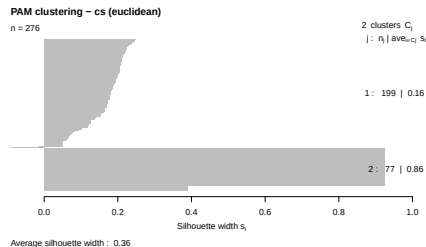
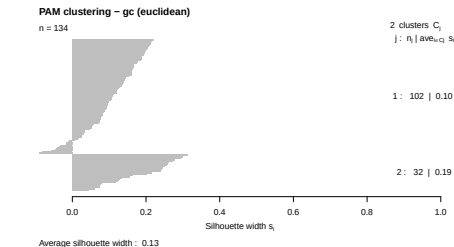
Backup — behavioural MDS (Manhattan)



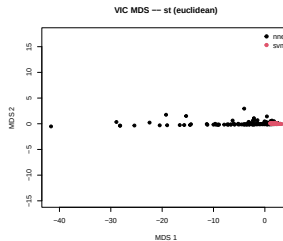
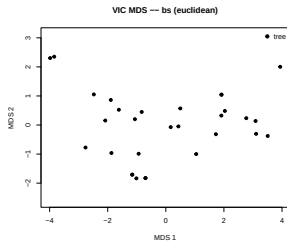
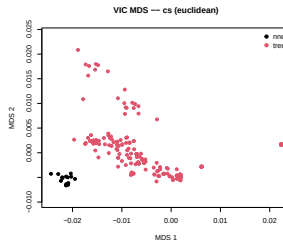
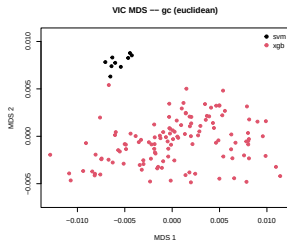
Backup — behavioural dendrograms (Euclidean)



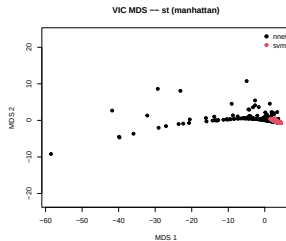
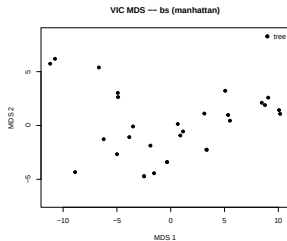
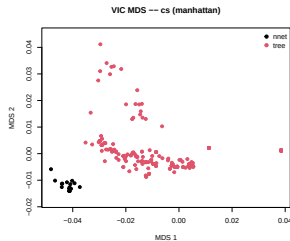
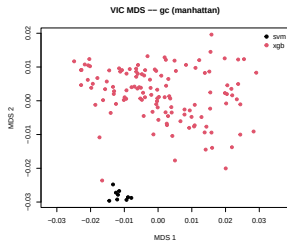
Backup — behavioural PAM, $k = 2$ (Euclidean)



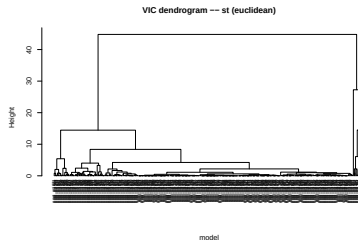
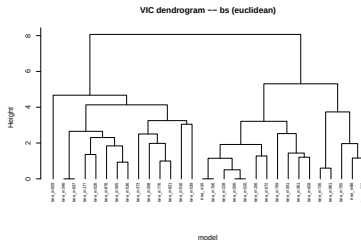
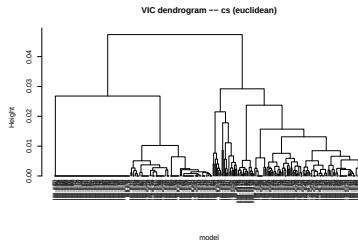
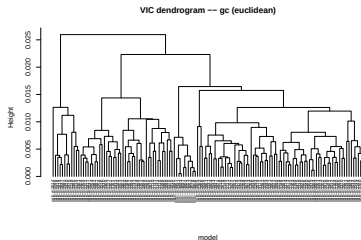
Backup — VIC MDS (Euclidean)



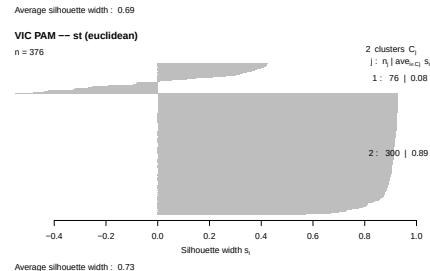
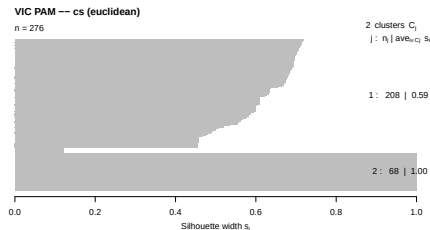
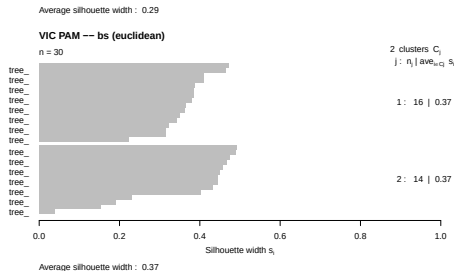
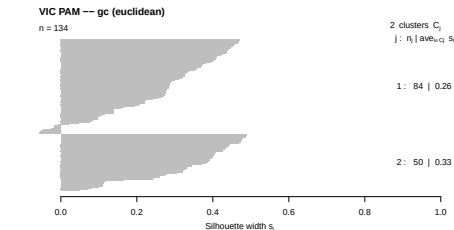
Backup — VIC MDS (Manhattan)



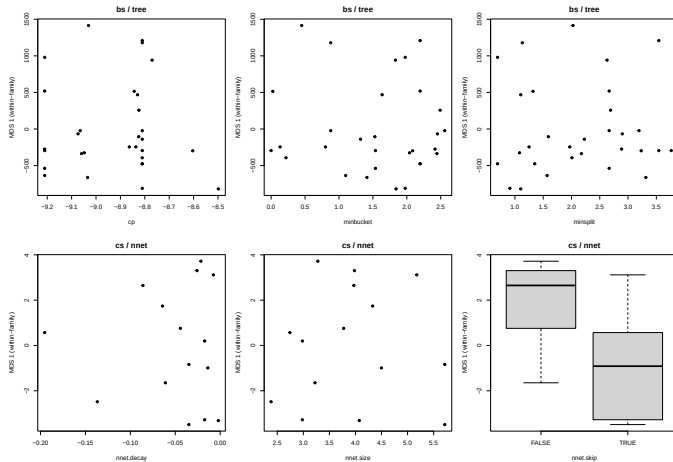
Backup — VIC dendrograms (Euclidean)



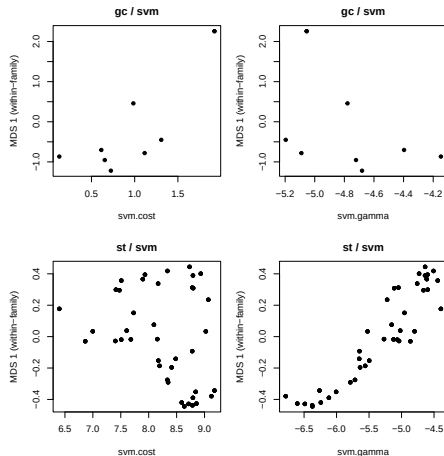
Backup — VIC PAM, $k = 2$ (Euclidean)



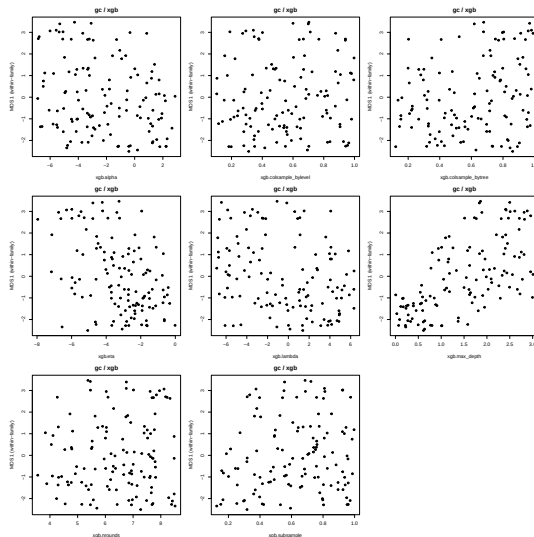
Backup — HP trajectories: bs/tree, cs/nnet



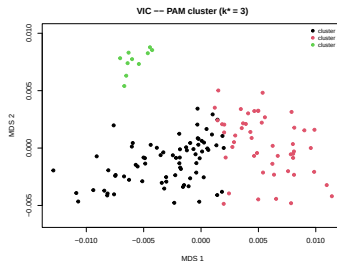
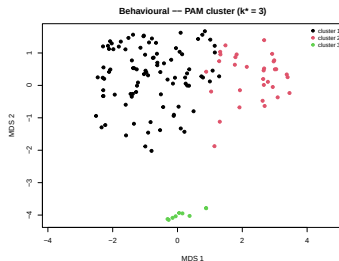
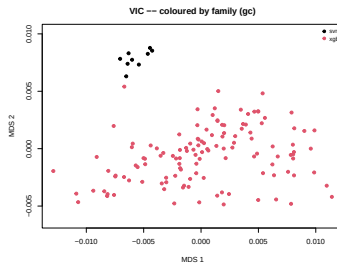
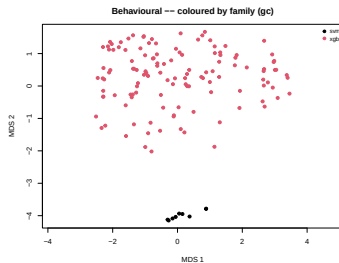
Backup — HP trajectories: gc/svm, st/svm



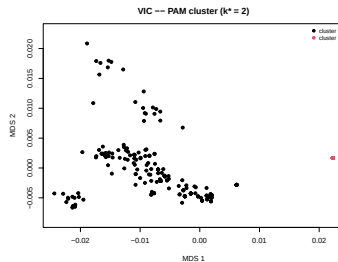
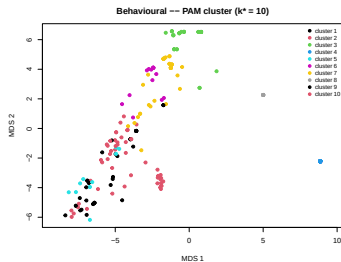
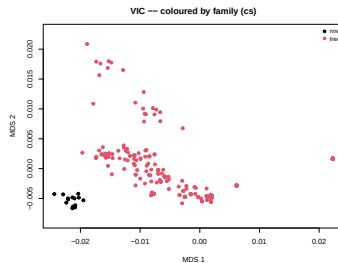
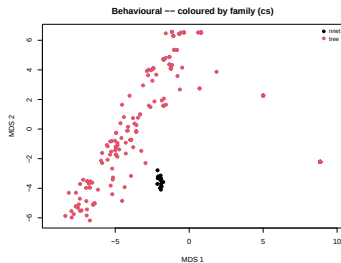
Backup — HP trajectories: gc/xgb



Backup — PAM cluster comparison gc



Backup — PAM cluster comparison cs



Backup — PAM cluster comparison bs

